

FixGrid Probe™: The Complete Ecosystem

A Live Web Platform (fixgrid.vytron.me) & A Frugal Handheld Diagnostic Wand Championing Sustainable Repair Across All 21+ Categories

Right-to-Repair & Circular Economy Platform

FixGrid Probe™

The Complete Ecosystem: A Live Web Platform (fixgrid.vytron.me) & A Frugal Handheld Diagnostic Wand Championing Sustainable Repair Across All 21+ Categories

✓ Live Digital Platform: Search & escrow across 21+ categories (Appliances, Tech, EVs)

✓ Physical Innovation: Under ₹1,800 FixGrid Probe™ sub-junction diagnostic wand

✓ Dual-Loop QR Warranty: Tamper-evident seals with photo proof

FIND A LOCAL REPAIR SHOP IN INDIA THAT CAN ACTUALLY FIX IT

Search vertical/horizontal repair shops and technicians across Delhi, Mumbai, Bangalore and top Indian cities. Compare diagnostics, see real-time operating hours, and book with warranty.

🔍 Classified across battery, motherboard, appliance service

SEARCH

Live Platform: fixgrid.vytron.me



FixGrid Probe™ Working Prototype

FixGrid Ecosystem • fixgrid.vytron.me

Live Web Platform + FixGrid Probe™ Diagnostics

Right-to-Repair & Circular Economy

KEY TAKEAWAYS & SLIDE CONTENT

- ✓ Live Digital Platform: Search & escrow across 21+ categories (Appliances, Tech, EVs)
- ✓ Physical Innovation: Under ₹1,800 FixGrid Probe™ sub-junction diagnostic wand
- ✓ Dual-Loop QR Warranty: Tamper-evident seals with photo proof

👤 PRESENTER SCRIPT & TALKING POINTS

"Welcome. FixGrid Probe unites a live web marketplace with a low-cost physical diagnostic wand, solving both the digital trust void and the technical hardware diagnostic barrier."

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SLIDE 01 OF 08

Problem Statement: The Throwaway Culture & Climate Paradox

Why 78% of repairable gadgets end up in landfills while local artisans lose their livelihoods

// 02 • PROBLEM STATEMENT

[02 / 08]

Problem Statement: The Throwaway Culture & Climate Paradox

1. The Trust Void Across All 21+ Categories

- 78% of consumers discard repairable goods (appliances, consumer tech, tools) fearing arbitrary pricing & zero warranty.
- Spans ALL 21+ categories: Major home appliances (ACs, fridges, washing machines), laptops, phones, TVs & EVs.
- Local repair artisans across every trade remain digitally invisible, informal, and overlooked.

2. The Universal Hardware Diagnostic Void

- Lab diagnostic equipment costs ₹50,000–₹1.5L—unaffordable for neighborhood appliance & tech workshops.
- Standard multimeters inject 2.5V–3.0V, risking delicate micro-controllers in modern smart appliances & boards.
- Without visual proof, mechanics guess—declaring entire appliance control boards or devices "unfixable."

₹1.2L Cr

THE LOST VALUE

Consumers waste billions replacing appliances & devices when an artisan 500m away could repair it.

85%+

SKILLED HEROES OVERLOOKED

Informal repair mechanics across 21+ categories lack digital tools and are trapped in unverified guesswork.

Millions

E-WASTE CATASTROPHE

Heavy appliances & electronics flood landfills, leaching toxic metals while factories burn carbon for replacements.

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Live Web Platform + FixGrid Probe™ Diagnostics

Right-to-Repair & Circular Economy

KEY TAKEAWAYS & SLIDE CONTENT

- ✓ The \$15.2B Repair Economy is Broken: 78% of consumers discard repairable smartphones, laptops, and appliances into landfills due to fear of arbitrary pricing and technician fraud.
- ✓ Hardware Diagnostic Barrier: Informal mechanics lack ₹1,50,000 lab oscilloscopes. Trial-and-error multimeters fry delicate microchips, misdiagnosing ₹10 capacitors as dead motherboards.
- ✓ Mounting E-Waste Crisis: Millions of tons of toxic e-waste generated annually, leaching lead and cadmium into groundwater while new manufacturing burns carbon.

👤 PRESENTER SCRIPT & TALKING POINTS

"Every day, thousands of repairable phones and appliances are thrown away. A ₹50 blown capacitor is declared a dead motherboard because roadside mechanics lack diagnostic tools. FixGrid ends this cycle."

A Two-Pillar Solution: Digital Marketplace + Smart Hardware

Unifying web discovery and escrow governance with hardware circuit precision

// 03 • PROPOSED SOLUTION

[03 / 08]

Proposed Solution: The FixGrid Hardware-Software Innovation

PILLAR 1: DIGITAL PLATFORM

FixGrid Trust & Warranty Cloud Engine

- All 21+ Categories: Covers Home Appliances, Consumer Tech, Power Tools, EVs & Wearables.
- Upfront Smart Escrow: Customer payments locked safely at booking; released on verified sign-off.
- 90-Day Digital Warranty: Platform-guaranteed warranty passport verifiable on any smartphone.
- Artisan Empowerment: Digital storefronts, verified reviews & 5% completed-job rebate.

PILLAR 2: HARDWARE WAND

FixGrid Probe™ Multimodal Diagnostic Tool

- 0.40V Sub-Junction Divider: Clamps below 0.6V P-N junction — tests in-circuit without desoldering.
- 16-Bit ADS1115 + Tone: Pitch-shift audio buzzer (1.2kHz → 2.4kHz) guides to shorted 0402 SMDs.
- ESP32-CAM Microscope: 2.5cm macro lens & shadowless ring LEDs captures HD fault photo proof.
- Hardware Trust-Gate: Transmits cryptographic VERIFIED_PASS token to release escrow funds.

₹1,795 BOM

Frugal Prototype Cost

0.40V Clamp

Sub-Junction In-Circuit

Dual-Loop QR

Digital Product Passport

Mission LiFE

E-Waste Mitigation Goal

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Live Web Platform + FixGrid Probe™ Diagnostics

Right-to-Repair & Circular Economy

KEY TAKEAWAYS & SLIDE CONTENT

- ✓ Pillar 1 — Live Web Platform (fixgrid.vytron.me): 21+ repair categories, upfront transparent pricing, safe escrow lock, and verified shop credentials.
- ✓ Pillar 2 — FixGrid Probe™ Multimodal Diagnostic Tool: 0.40V sub-junction divider testing in-circuit without desoldering, 16-bit ADS1115 audio tone buzzer, and ESP32-CAM microscope.
- ✓ Key Innovation Metrics: ₹1,795 BOM prototype cost, 0.40V safe voltage clamp, Dual-Loop QR digital product passport, supporting Mission LiFE.

PRESENTER SCRIPT & TALKING POINTS

"FixGrid Probe merges physics and digital governance. It tests in-circuit at 0.40V without desoldering, snaps macro photos of 0402 SMDs, links to a Dual-Loop QR warranty seal, and acts as a hardware trust-gate releasing escrow."

FixGrid Probe™: Lab Diagnostics in a Handheld Wand

Sub-junction in-circuit fault detection with 5MP macro photo evidence under ₹1,800 BOM

// 04 • HARDWARE INNOVATION

[04 / 08]

FixGrid Probe™: Lab Diagnostics in a Handheld Wand



FixGrid Probe™ Working Prototype (₹1,795 / ~\$21 BOM)

MICRO-TOUCH SENSING

Instant Fault Chirp

Technician touches probe needle to circuit trace. Conducts rapid impedance testing in milliseconds, beeping and lighting Green (Normal) or Red (Fault) on OLED screen. Safe sub-3.3V (<5mA) sensing prevents static shock.

MAGNIFIED VISUALS

5MP Macro Photo Proof

Integrated 5MP macro camera with polarized ring LEDs captures 10x close-up photos of burned ICs or cracked solder joints, sending undeniable visual proof straight to the customer's smartphone before soldering.

DIGITAL PASSPORT

Dual-Loop QR Warranty

Tool writes a physical tamper-evident QR void seal placed over device or appliance seam. Customer scans to review diagnostic readings, before/after photos, and activates a guaranteed 90-day digital warranty.

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Live Web Platform + FixGrid Probe™ Diagnostics

Right-to-Repair & Circular Economy

KEY TAKEAWAYS & SLIDE CONTENT

- ✓ Instant Fault Chirp (Micro-Touch Sensing): Rapid impedance testing in milliseconds; beeps and glows Green (Normal) or Red (Fault) on OLED screen without desoldering.
- ✓ 5MP Macro Photo Proof (Magnified Visuals): Integrated macro camera with polarized ring LEDs captures 10x close-up photos of burned ICs or cracked joints, beaming undeniable proof straight to the customer's phone.
- ✓ Dual-Loop QR Warranty (Digital Passport): Writes a physical tamper-evident holographic QR void seal over the chassis seam, locking in a guaranteed 90-day digital warranty.

👤 PRESENTER SCRIPT & TALKING POINTS

"FixGrid Probe provides three capabilities in one wand: micro-touch fault sensing with instant beeps, macro photo proof sent to the phone, and a tamper-evident digital warranty seal."

fixgrid.vytron.me: Consumer Marketplace & Trust Engine

Connecting device owners to verified neighborhood repair artisans with escrow safety

// 05 • DIGITAL PLATFORM

[05 / 08]

FixGrid Web Platform (fixgrid.vytron.me)

Hyper-Local Search & Map Engine

Allows consumers to discover verified repair shops within walking distance across 21+ categories (Major Home Appliances, ACs, Microwaves, Phones, Laptops, Power Tools, EVs) with live status.

- Next.js 16 + Leaflet Map Engine

Smart Escrow & Payment Protection

Customer payments are locked securely on the platform upon booking and released to the shopkeeper only after the customer inspects and signs off on verified satisfaction.

- Supabase PostgreSQL Triggers & RLS

Artisan Storefronts & 5% Cashback

Provides informal repair mechanics with instant SEO-indexed digital profiles, verified reviews, diagnostic log uploads, and a 5% completed-bill cashback incentive.

- Shopkeeper Empowerment Engine

The screenshot displays the FixGrid web platform interface. At the top, it says 'FIND A LOCAL REPAIR SHOP IN INDIA THAT CAN ACTUALLY FIX IT'. Below this, there's a search bar with the text 'Cracked screen, battery replacement, appliance service' and a 'SEARCH' button. The interface also shows a map of India with a location pin. At the bottom, it says 'Live Production Platform at fixgrid.vytron.me'.

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Live Web Platform + FixGrid Probe™ Diagnostics

Right-to-Repair & Circular Economy

KEY TAKEAWAYS & SLIDE CONTENT

- ✓ 21+ Repair Categories: Smartphones, laptops, home appliances, microwaves, smartwatches, and clean-tech EV battery chargers.
- ✓ Smart Escrow Safety: 100% upfront payment held securely; released to technician only after hardware test pass and customer sign-off.
- ✓ Verified Local Directory: Interactive Leaflet map engine finding skilled artisans within 500m to 5km with genuine storefront audits.

👤 PRESENTER SCRIPT & TALKING POINTS

"Our live web platform fixgrid.vytron.me is operational with interactive maps, verified expert profiles, upfront pricing, and smart escrow protection across 21 categories."

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SLIDE 05 OF 08

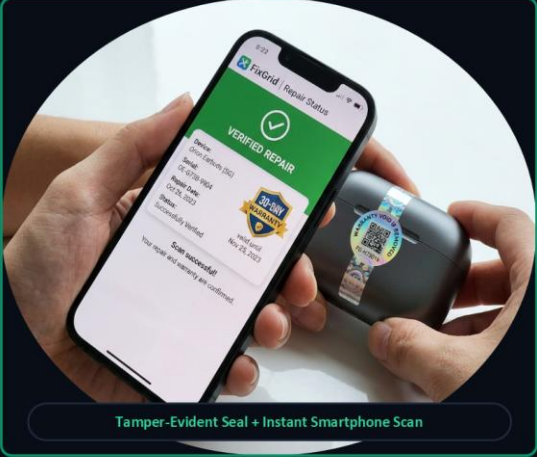
Physical & Digital Synergy: The Dual-Loop Warranty

How hardware diagnostics, physical seals, and cloud telemetry ensure 100% accountability

// 06 • HARDWARE-SOFTWARE BRIDGE

[06 / 08]

Dual-Loop QR Digital Product Passport



Tamper-Evident Seal + Instant Smartphone Scan

STEP 01

FixGrid Probe Diagnostics Upload

When the technician completes the repair across any category, the FixGrid Probe wand syncs the passing impedance test and high-res macro photos directly to the FixGrid cloud database via ESP32 Wi-Fi/BLE.

STEP 02

Physical Holographic Void Seal

A serialized, tamper-evident holographic QR void sticker is printed and affixed over the device or appliance seam. If anyone attempts to tamper with or reopen the unit, the physical pattern breaks.

STEP 03

Customer Scan & Warranty Activation

The customer scans the QR code with any standard smartphone camera. It displays component before/after photos, diagnostic health logs, and activates an ironclad 90-day digital warranty certificate.

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Live Web Platform + FixGrid Probe™ Diagnostics

Right-to-Repair & Circular Economy

KEY TAKEAWAYS & SLIDE CONTENT

- ✓ Step 01 — FixGrid Probe Diagnostics Upload: Technician finishes repair; wand syncs passing impedance telemetry and macro photos directly to FixGrid cloud via ESP32 Wi-Fi/BLE.
- ✓ Step 02 — Physical Holographic Void Seal: Serialized tamper-evident QR void sticker is affixed over the device seam. If tampered with, the pattern breaks.
- ✓ Step 03 — Customer Smartphone Scan & Warranty Activation: Customer scans QR with any phone camera to view before/after photos and activate an ironclad 90-day warranty certificate.

👤 PRESENTER SCRIPT & TALKING POINTS

"This is where the magic happens: the FixGrid Probe hardware records the repair, generates a tamper-evident holographic QR seal, and the customer scans it on their phone to see verified before-and-after photos and activate their warranty."

Tapping the \$15.2B Repair Economy

Sustainable monetization that aligns consumer savings, artisan livelihoods, and e-waste reduction

// 07 • IMPACT & SCALABILITY

[07 / 08]

Tapping the \$15.2B Repair Economy

60–80%**ECONOMIC (Consumer & Artisan)**

Consumers save 60–80% compared to purchasing new devices. Local mechanics gain formal digital identity, repeat customers, and higher daily earnings.

100%**SOCIAL (Community Dignity)**

Brings unorganized street artisans into the formal digital economy, elevating informal mechanics with lab-grade diagnostic tools and transparent trust.

Zero**ECOLOGICAL (Planet Circularity)**

Diverts hundreds of kilograms of toxic printed circuit boards and lithium batteries from municipal landfills, cutting manufacturing carbon emissions.

1. Platform Take-Rate (5–8%)

Competitive transaction fee on every escrow-verified repair booking.

2. Customer Care+ Subscription (₹199/mo | ₹1,499/yr)

Household repair pass: free FixGrid Probe diagnostics, zero booking fees & 60-day warranty.

3. Shop Pro SaaS & Hardware Tier (₹999/mo)

FixGrid Probe firmware updates, warranty management, digital store CRM & analytics.

4. Certified Spare Parts Logistics

Curated B2B marketplace revenue for verified authentic micro-components.

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Live Web Platform + FixGrid Probe™ Diagnostics

Right-to-Repair & Circular Economy

KEY TAKEAWAYS & SLIDE CONTENT

- ✓ Triple Bottom Line: Consumers save 60–80% vs buying new; local artisans gain respected digital identity; tons of hazardous e-waste prevented at the source.
- ✓ Stream 1 — Platform Take-Rate: Competitive 5–8% transaction fee on every escrow-verified repair booking.
- ✓ Stream 2 — Customer Care+ Pass: ₹199/month household subscription with free FixGrid Probe diagnostics & zero booking fees.
- ✓ Stream 3 — Shop Pro SaaS: ₹999/month repair shop tier for firmware updates, warranty management, and digital CRM.
- ✓ Stream 4 — B2B Spare Parts: Curated marketplace logistics for verified authentic micro-components.

🗣️ PRESENTER SCRIPT & TALKING POINTS

"Our model creates value for everyone: consumers save money, informal mechanics earn reliable livelihoods, and FixGrid monetizes through transaction fees and subscriptions."

A Movement Towards a Sustainable Repair Economy

Phased scaling from neighborhood repair clusters to nationwide municipal partnerships

// 08 • THE FUTURE ROADMAP

[08 / 08]

A Movement Towards a Sustainable Repair Economy

01

PHASE 1 (MONTHS 1–6)

Pilot Repair Clusters

- Deploy 50 FixGrid Probe units across Mumbai electronics repair markets.
- Validate web escrow & before/after macro photo flow with 1,000+ customer repairs.

02

PHASE 2 (MONTHS 7–12)

Community Drives

- Partner with schools & colleges for zero-e-waste community repair clinics.
- Roll out Customer Care+ subscription pass and certified technician onboarding.

03

PHASE 3 (YEAR 2+)

Municipal Scaling

- Integrate with municipal e-waste boards for certified gadget triage.
- Scale frugal FixGrid Probe tool manufacturing to local repairers nationwide.

FixGrid Probe™ Unified Ecosystem

A Unified Web & Hardware Solution for Right-to-Repair, Consumer Trust & Circular Sustainability

Project FixGrid

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Live Web Platform + FixGrid Probe™ Diagnostics

Right-to-Repair & Circular Economy

KEY TAKEAWAYS & SLIDE CONTENT

- ✓ Phase 1 (Months 1–6) — Pilot Repair Clusters: Deploy 50 FixGrid Probe units across Mumbai electronics markets; validate web escrow with 1,000+ customer repairs.
- ✓ Phase 2 (Months 7–12) — Community Drives: Partner with schools and colleges for zero-e-waste community repair clinics; expand Care+ subscription.
- ✓ Phase 3 (Year 2+) — Municipal Scaling: Integrate with municipal e-waste recycling boards for certified gadget triage; scale frugal hardware tool manufacturing nationwide.
- ✓ Unified Vision: A world where repair is as trusted, easy, and respected as buying new.

🗣️ PRESENTER SCRIPT & TALKING POINTS

"Thank you. FixGrid and FixGrid Probe demonstrate that with frugal engineering and digital trust, we can empower local artisans, save consumers money, and protect our environment from toxic e-waste."